UNSUPERVISED LEARNING FOR CONSUMER FINANCE SEGMENTATION

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1. EXECUTIVE SUMMARY & DATA ARCHITECTURE

This report investigates consumer financial structures using data from the Federal Reserve Board's Survey of Consumer Finances (SCF). Unsupervised machine learning models—specifically K-Means Clustering and Principal Component Analysis (PCA)—are implemented to segment US households based on asset allocation, liabilities, net worth, and risk tolerance. Financial institutions utilize segmentation to craft tailored credit products, wealth management services, and risk assessment frameworks.

1.1 Survey of Consumer Finances (SCF) Overview

The SCF dataset provides micro-level cross-sectional data on household balance sheets. The survey includes high-dimensional financial variables measured in standardized dollar values and ordinal risk scales. Due to severe right-skewness in wealth distributions, non-linear variance-threshold pruning and robust scaling are necessary pre-processing steps prior to clustering.

VARIABLE	DESCRIPTION	FINANCIAL DOMAIN	DATA TYPE
NETWORTH	Total household net worth (Assets minus Liabilities)	Overall Wealth	Continuous ($)
ASSET	Total value of all financial and non-financial assets held	Asset Base	Continuous ($)
DEBT	Total outstanding household debt obligations	Liabilities	Continuous ($)
FIN / NFIN	Financial assets vs. Non-financial assets (Real estate, vehicle)	Asset Mix	Continuous ($)
HOUSES / RESNT	Value of primary residence vs. secondary real estate holdings	Real Estate	Continuous ($)
BUS / KGBUS	Total business equity and unrealized capital gains on business	Enterprise Ownership	Continuous ($)
INCOME	Total annual gross household income	Cash Flow	Continuous ($)
RISK	Willingness to take financial risk (Scale 1: Refuse to 4: Substantial)	Psychometric Risk	Ordinal Scale

2. HIGH-DIMENSIONAL FEATURE SELECTION & VARIANCE PRUNING

Clustering algorithms based on Euclidean distance metrics suffer from the *curiosity of dimensionality*, where high-dimensional space causes distance measures to become uniform. To isolate informative features, we evaluate feature variance across the dataset.

2.1 Mathematical Formulation of Feature Variance

For a given feature column $\mathbf{x}$ across N household observations with sample mean $\bar{\mathbf{x}}$, the sample variance s^2 is computed as:

$$s^2 = (1 / (N - 1)) \times \sum_{i=1}^{N} (x_i - \bar{x})^2$$

Because financial values span several orders of magnitude (from $0 to tens of millions), raw dollar variance is dominated by large-scale balances (e.g., **NETWORTH**). Features are scaled via Z-score standardization prior to variance filtering:

$$z_i = (x_i - \bar{x}) / s$$

FEATURE SELECTION STRATEGY

Features exhibiting standard variance below a threshold ($Var(z) < 0.10$) are pruned. The top features retained for clustering represent household leverage (**DEBT/ASSET**), liquidity ratio (**FIN/ASSET**), real estate allocation, and risk aversion quotient.

3. K-MEANS CLUSTERING ALGORITHM & MODEL CALIBRATION

The K-Means algorithm partitions N household observations into K distinct, non-overlapping clusters. Each household is assigned to the cluster with the nearest centroid mean.

3.1 Mathematical Objective: Inertia (WCSS)

The objective function minimizes the Within-Cluster Sum of Squares (WCSS), also known as model inertia:

$$Inertia\ (J) = \sum_{k=1}^{K} \sum_{i \in C_k} || x_i - \mu_k ||^2$$

Where C_k represents the set of household observations in cluster k, and μ_k is the mean vector (centroid) of cluster k.

3.2 Optimal Cluster Determination (Elbow Method)

To determine the optimal value of K, we evaluate inertia across candidate clusters from $K = 1$ to $K = 10$. The point of diminishing marginal returns (the "elbow") indicates the optimal trade-off between model simplicity and variance explanation.

NUMBER OF CLUSTERS (K)	INERTIA (WCSS)	ABSOLUTE REDUCTION	RELATIVE INERTIA CHANGE (%)
K = 1	142,500.00	—	Baseline (100%)
K = 2	88,200.50	54,299.50	−38.1%
K = 3	52,140.20	36,060.30	−40.8%
K = 4 (Optimal)	**34,800.80**	**17,339.40**	**−33.2%**
K = 5	29,100.10	5,700.70	−16.3%
K = 6	25,400.40	3,699.70	−12.7%

3.3 Household Segment Profiles (K = 4 Centroid Analysis)

Segmenting the dataset into four clusters yields distinct, actionable consumer finance profiles:

CLUSTER LABEL	HOUSEHOLD SEGMENT NAME	MEAN NET WORTH ($)	DEBT-TO-ASSET RATIO	PRIMARY ASSET DRIVER	RISK PROFILE
Cluster 0	Moderate Middle Class	$185,000	0.42	Primary Residence (HOUSES)	Low-to-Moderate
Cluster 1	High Net Worth Investors	$2,450,000	0.08	Equity & Financial (FIN)	Substantial Risk
Cluster 2	Highly Leveraged Debtors	$42,000	0.78	Unsecured Debt / Installments	High Risk / Low Cap
Cluster 3	Business Owners & Entrepreneurs	$1,820,000	0.24	Business Equity (BUS)	Above Average Risk

4. DIMENSIONALITY REDUCTION VIA PRINCIPAL COMPONENT ANALYSIS

To visualize the cluster partitions and understand latent structural drivers, Principal Component Analysis (PCA) is applied to project the feature space onto orthogonal linear axes.

4.1 Mathematical Framework of PCA

Given a standardized data matrix **Z** of dimension $N \times p$, the sample covariance matrix $\sum$ is calculated as:

$$\sum = (1 / (N - 1)) \times Z^T Z$$

Eigen-decomposition yields eigenvalues λ_j and eigenvectors v_j satisfying:

$$\sum v_j = \lambda_j v_j$$

The total variance captured by the m-th principal component is given by the proportion of its eigenvalue relative to the sum of all eigenvalues:

$$\textit{Explained Variance Ratio } (PC_m) = \lambda_m / (\sum_{j=1}^{p} \lambda_j)$$

4.2 Principal Component Loading & Variance Analysis

COMPONENT	EIGENVALUE (Λ)	INDIVIDUAL VARIANCE (%)	CUMULATIVE VARIANCE (%)	DOMINANT CONTRIBUTING FEATURES
PC 1	3.82	38.2%	38.2%	NETWORTH (+), ASSET (+), FIN (+)
PC 2	2.14	21.4%	59.6%	DEBT (+), LEVERAGE (+), RISK (+)
PC 3	1.35	13.5%	73.1%	BUS (+), KGBUS (+), NFIN (+)

COMPONENT	EIGENVALUE (Λ)	INDIVIDUAL VARIANCE (%)	CUMULATIVE VARIANCE (%)	DOMINANT CONTRIBUTING FEATURES
PC 4	0.91	9.1%	82.2%	HOUSES (+), RESNT (−)

4.3 Latent Structural Dimensions

- **PC 1 (Wealth Accumulation Axis):** Captures overall household solvency and liquid capital holdings. High positive scores correlate with affluent institutional investors.
- **PC 2 (Credit Risk & Leverage Axis):** Captures outstanding liabilities scaled by total asset base. High scores identify vulnerable or over-leveraged households.
- **PC 3 (Entrepreneurial Capital Axis):** Separates passive wage earners from active private enterprise operators.

STRATEGIC IMPLICATIONS FOR CONSUMER FINANCE

By transforming customer records into 2D spaces governed by PC1 and PC2, financial institutions can automate risk scoring, set targeted credit limits, and deploy customized wealth preservation or debt consolidation programs.